3rd SEMESTER INTERNSHIP REPORT ON

Analyzing the Dynamics of Urban Flooding in Guwahati: Causes, Impacts, and the Efficacy of Management Strategies (2024-2025)

SUBMITTED TO

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE FROM

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

This is to certify that the 3rd semester internship report entitled

"Analyzing the Dynamics of Urban Flooding in Guwahati: Causes, Impacts, and the Efficacy of Management Strategies (2024-2025)"

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Sincerely,

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Abstract

Guwahati, the largest city of Assam, faces the problem of floods and waterlogging almost every year. The issue has become more serious in recent times because of very heavy rainfall, hilly runoff, poor drainage, and the shrinking of natural wetlands. During our internship, we tried to study this issue to understand the main causes, its impacts on the people of the city, and the steps taken by the authorities.

We observed that floods affect almost every part of life in the city—transport, houses, schools, businesses, and even health. Projects like Mission Flood-Free Guwahati have started, but problems such as clogged drains, encroachment on wetlands, and lack of awareness among the public are still present.

From this study, we concluded that Guwahati needs a proper long-term drainage plan, protection of its wetlands, regular cleaning of drains, better waste management, and more involvement of citizens. During our internship, we understood that these steps are not only the duty of the government but also require awareness and support from the people.

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Introduction

Guwahati is the capital city of Assam and the gateway to Northeast India. It is also one of the fastest growing cities in the region. Because of this rapid growth, the city is facing many urban challenges, and one of the biggest among them is flooding.

Every year, especially during the rainy season from June to September, different parts of Guwahati go under water. People are unable to move, schools and offices remain closed, vehicles get stuck, and life becomes difficult. In 2024 and 2025, this problem was seen in an even bigger way, as the city recorded very heavy rainfall.

As part of our 3rd semester internship, we studied this issue by reading reports, following news articles, and observing officials and local people. The purpose of this report is to share what we learned in a simple way, so that others can also understand why Guwahati floods every year and what can be done about it.

Background and Current Situation (2024-2025)

Flooding in Guwahati is not new. But in recent years, the scale of the problem has increased. The city lies in a bowl-shaped valley surrounded by small hills. Water from these hills, combined with heavy rainfall quickly collects in the low areas of the city.

In May 2025, Guwahati recorded 111 mm of rainfall in just 24 hours. This broke past records and created flash floods in many areas like Chandmari, Hatigaon, Beltola, Rukminigaon, Zoo Road, and GS Road. Vehicles were stranded, many shops and houses were flooded, and rescue teams had to use rubber boats to evacuate people.

Even in June 2025, heavy rain caused floods in almost 18 municipal wards, affecting thousands of residents. Power supply was interrupted in some areas, and transportation through roads and ferries was badly hit. These events show that despite government measures, the problem continues to grow and needs urgent attention.

Causes of Urban Flooding in Guwahati

From our study, we found that the flooding problem is caused by both natural and human-made reasons.

A. Natural Causes:

- **Heavy Rainfall:** Guwahati receives very high rainfall in the monsoon, and sometimes even during pre-monsoon showers. When too much rain falls in a short time the drains cannot carry it away.

Heavy Monsoon Rainfall

- **Bowl-Shaped Topography:** Since the city lies in a low valley surrounded by hills, water flows down quickly and collects in the city.

- **Rivers and Streams:** The Bharalu, Bahini, Basistha, and other streams swell up during rain and overflow into nearby areas.

B. Human-Made causes:

- **Unplanned Urbanization:** Guwahati has grown very fast. Wetlands like Silsako Beel and parts of Deepor Beel are being filled up for construction. These wetlands earlier worked like storage areas for rainwater but now they cannot serve this purpose.

Unplanned Urban Development

- **Poor Drainage System:** Most of the drains are small, shallow and often blocked with plastic, soil and waste. When drains are clogged, even normal rain causes waterlogging.

Overflowing drains

- **Construction Projects:** Many new flyovers and road projects do not have proper drainage planning. During our internship, we also came across complaints from local people who said that rainwater from flyovers directly enters their homes.

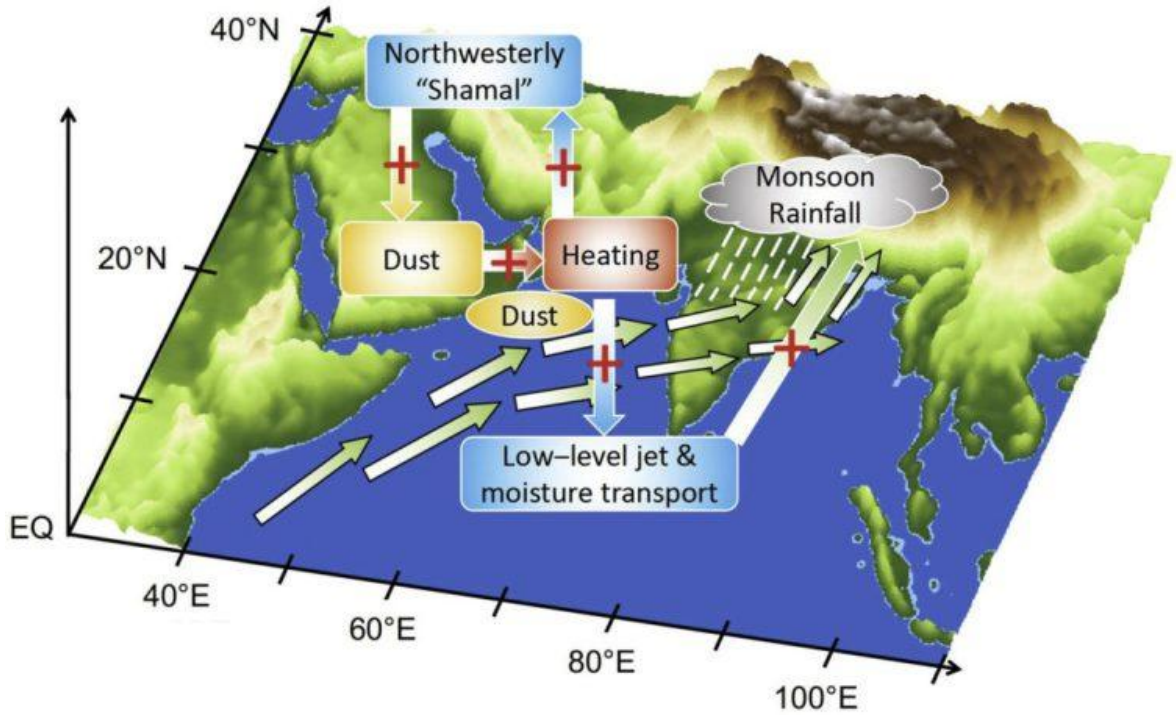
- **Old Infrastructure:** Many of the older culverts and drains have not been replaced or repaired. These outdated structures fail during heavy rain.

- **Lack of proper planning:** Guwahati does not have a single long-term drainage plan for the whole city. Because of this, only small local solutions are made here and there and they do not solve the flooding problem fully.

- **Water Release from Dams:** Sometimes sudden release of water from upstream dams raises water levels in streams which adds to the flooding problem.

C. Climate Change:

- Climate change is one of the main reasons why floods in Assam are becoming worse. It is making rainfall more irregular and sometimes heavier than before. Because of this, flood periods are also becoming longer. For example in 2022, Assam faced floods for almost seven months.
- Scientists have noticed that in Northeast India, the southwest monsoon is arriving earlier, even in May. This brings heavy rainfall in a shorter period which increases flooding.
- Assam is also considered one of the most flood-prone regions in the world. Studies say that by 2050, the state's risk of climate-related damage will increase more than three times compared to 1990, which shows how vulnerable the region is.



Source: [Earth-Science Reviews/Science Direct]

Impacts of Urban Flooding in Guwahati

The flooding has many negative effects on the city and its people-

- **Daily Life:** Floods stop normal life. People cannot travel, traffic jams become common, buses and ferries stop and schools are often closed.

Disrupted Transportation(Source:thehillstimes.in)

- **Infrastructure Damage:** Roads break apart, houses get filled with water and shops lose their goods. Repairing this damage costs a lot of money.

- **Health Hazards:** Flood water becomes a breeding ground for mosquitoes. Diseases like dengue and malaria increase. Contaminated water also spreads stomach infections. In 2025, there was a rise in such diseases.

- **Economic Losses:** Small shops and businesses face the most loss. When customers cannot come and goods are destroyed, the economy suffers.

- **Stress on People:** People live in fear whenever there is heavy rain. Students miss classes and elderly citizens find it difficult to move in waterlogged areas.

- **Environmental Impact:** Wetlands are destroyed and dirty floodwater pollutes the city environment.

Schools Closed due to Floods

Flood Management Strategies and Challenges in Guwahati (2024-2025)

The government and civic bodies have taken some steps, but they are not enough. From our study, we identified the following strategies and challenges:

- **Master Drainage Plan:** A proper city-wide drainage plan is needed. At present, only small local solutions are made, which are not effective.

- **Wetland Protection:** Wetlands like Silsako Beel must be restored and protected. This will help in storing extra rainwater.

- **Drainage Maintenance:** Drains must be cleaned regularly, not just before the monsoon. Machines for desiltation should be used more often.

Drain Desiltation Work (Photo: @DCKamrupMetro / X):

- **Waste Management:** Plastic and garbage must not enter the drains. Public awareness campaigns should be run to teach people not to throw waste in drains.

- **Climate-Safe Planning:** New roads and buildings should be made in a way that allows water to drain. Green methods like rain gardens and permeable pavements can be used.

- **Better Construction Rules:** Flyovers and big projects should have mandatory drainage provisions. Silt from construction sites should be controlled.

- **Coordination:** Different government departments must work together. We learned that many projects get delayed because there is no proper communication between them.

- **Community Role:** People should take part by keeping their surroundings clean and avoiding illegal construction on wetlands.

- **Use of Technology:** Tools like GIS maps and flood warning systems should be used to forecast and manage floods better.

Silsako Beel Wetland Restoration

Conclusion

From our internship study, we understood that urban flooding in Guwahati is caused by both natural and man-made reasons. The city's bowl-shaped location and heavy rainfall make it naturally prone to floods but unplanned growth, encroachment on wetlands and poor drainage have made the problem much worse.

Although projects like Mission Flood-Free Guwahati are going on, the recurring floods of 2024 and 2025 show that much more needs to be done. We believe that a proper drainage master plan, protection of wetlands, strict construction rules, better waste management and active participation from the people are all necessary.

Through this internship study, we learned that both natural reasons and human mistakes are making the city's flooding problem worse. We also understood how important proper planning, public awareness and responsibility from everyone are if the city is to become safer in the future.

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